



Fig. 2: E-2C Hawkeye, AEW aircraft (top); F-18 combat aircraft in a typical strike load out (centre to wingtip): external fuel tanks, bombs, anti-radiation missiles and short-range air-to-air missiles (bottom)

same time, come with a sky-high bill. For CSGs to survive and unleash their power, various electronic systems serve as the backbone. The electronic systems present in USN's CSGs and how these play crucial roles are the subject of foci in this mil-tech article.

Defensive combat operations

The purpose of a CSG is power projection; in simple words, launching combat aircrafts from the carrier stationed at deep seas and bombing the enemy. But the carrier being a highly-praised and priced target, objective of the enemy is to first destroy the carrier when the war begins. This could even turn the tide of the war.

Being a gigantic sitting duck on the sea, the enemy can easily find the carrier and attack either using aircraft, surface ships or submarines. So, in order to protect the carrier, a CSG typically comprises four to six destroyers (a kind of warship), two cruisers (a larger warship), and a nuclear attack submarine.

Destroyers are highly specialised in defending the carrier from threats posed by enemy surface ships and submarines. These also have respectable capability against aircrafts.

Cruisers, which are slightly larger

than destroyers, are specialised in taking care of enemy air threats. Due to a larger size, these possess more weapon capability than destroyers. While destroyers operate at around 40km from the carrier to form a defensive perimeter against surface and sub-surface threats, cruisers operate in the vicinity of the carrier to provide air defence.

The two complement each other. Through a myriad of sensors, these can detect and neutralise the threats arising from all directions. Nuclear submarines sail well-ahead of the carriers looking for enemy submarines and operate loosely bound to the carrier. All these vessels try to form an impenetrable virtual bubble around the carrier.

A CSG wages offensive air strike battles (strike warfare) on the enemy and, at the same time, wages defensive battles against enemy aircrafts, enemy submarines and enemy surface vessels.

Air defence warfare

During World War II, in Battle of Midway, a group of aircrafts from the USN came directly above IJN carriers Soryu, Kaga and Akagi, and bombed these. This was the turning point of

War at Pacific and even of World War II. Losing these carriers in just minutes proved too crippling for Japan, which ultimately put an end to Japanese Empire's naval supremacy in the Pacific in the days that followed. This ultimately led to the defeat of the Empire of Japan itself.

In hindsight, the radar's absence in Japanese carriers was the key. Ship-borne radars were yet to be introduced in these carriers. So, American aircrafts were able to bomb Japanese carriers undetected. But American carriers had been fitted with ship-borne radars, using which these could detect Japanese aircrafts beforehand and launch their aircrafts to defend in the nick of time.

Now, desperate Japan was looking to neutralise American carriers. To do so, some Japanese pilots rammed their aircrafts, fully-loaded with bombs and fuel, into American carriers, in what is called Kamikaze Attack. All these modes of attacks served as lessons for USN. When it started its ambitious carrier building and operating programme, air defence was given utmost priority.

Further to cement the importance of air defence, the USSR (of which the present Soviet Union was a part) entered into the picture as the newfound adversary of the USA to fight Cold War, a war fought without pulling a shot. The USSR sensed the challenging supremacy of the American carriers and fielded the deadly adversary of the carriers, the cruise missiles.

Having been developed inspired by Kamikaze aircrafts, cruise missiles offered a convenient means to attack a carrier. Earlier cruise missiles were just unmanned Kamikaze aircrafts.

From the cost perspective, a cruise missile costs a fraction of a carrier. But it is a very viable weapon, capable of being launched from bomber aircraft, warships and submarines. Incidentally, many of the early Soviet cruise missiles had nuclear warheads, sufficient enough to melt the entire steel present in the CSG.